Foundation Model-Based Multimodal Remote Sensing Data Classification

Xin He, Yushi Chen, *Member, IEEE*, Lingbo Huang, Danfeng Hong, *Senior Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— With the increasing availability and openness of remote sensing (RS) data collected from diverse sensors, there has been a growing interest in multimodal RS data classification. Nowadays, in the area of deep learning, there is a paradigm shift with the rise of foundation models, which are trained on large-scale datasets and are adaptable to a wide range of downstream tasks. In this study, the potential and effectiveness of foundation models for multimodal RS data classification is investigated. The training datasets of foundation models and multimodal RS datasets are quite different, and therefore, it is difficult to use a pretrained foundation model for multimodal RS data classification directly. To mitigate this difficulty, this article proposes a foundation model adaptation (FMA) framework for multimodal RS data classification without fine-tuning the parameters. Specifically, two learnable modules, i.e., cross-spatial interaction module and cross-channel interaction module, are proposed to add to the foundation model for extracting multimodal-specific representations. The cross-spatial and cross-channel interaction modules extract the characteristics of unimodal features along the spatial dimension and channel dimension, respectively. To effectively tackle the disparities among various RS modalities, an alignment approach (FMA2) is further explored based on the FMA. The FMA2 describes dependencies between different modalities by establishing a coupling score function, which can further enhance classification performance. To demonstrate the effectiveness and superiority of the FMA framework, comprehensive experiments are conducted on three multimodal RS datasets, showing improvement over the advanced multimodal RS data classification image methods.**

***Index Terms*— Foundation model, image classification, multimodal data, remote sensing (RS), transformer.**

I. Introduction

REMOTE sensing (RS), which observes objects from a distance without being in direct contact with them, has become the core technique for Earth observation tasks. The obtained ever-growing RS data, including hyperspectral image (HSI) [1], light detection and ranging (LiDAR) [2], and synthetic aperture radar (SAR) [3] data, have been applied to land cover classification, mineral exploration and mapping, environmental monitoring [4], urban planning [5], and military reconnaissance [6]. The above variety of applications is related to RS classification, which is a challenging data processing task [7].

In multimodal RS data classification, each type of RS data provide unique representation of an imaged scene [8]. For example, HSI captures information across a wide range of contiguous spectral bands, providing detailed spectral signatures of the Earth's surface [9]. This modality enables the identification and characterization of materials based on their unique spectral responses [10]. On the other hand, LiDAR uses laser pulses to measure the distance between the sensor and the Earth's surface. This modality provides information about the elevation, structure, and height of objects [11]. While SAR is not affected by weather conditions, such as clouds, fog, or darkness, it can operate effectively day and night, making it a reliable choice for continuous monitoring and data collection, regardless of weather conditions [12]. Optical RS data have relatively high spatial resolution, allowing for detailed characterization of land cover features [13].

Different RS data have variations in spatial resolution, spectral range, and sensitivity to different ground materials. Joint exploitation of multimodal RS data is capable of providing complementary information of the same scene, and therefore, opening a new way for multimodal RS data classification. For instance, the materials sharing similar spectral properties can hardly be separated by using only HSI [14]. By contrast, LiDAR gives depth and intensity information about the ground elevation, and SAR offers amplitude and phase geometrical information about the Earth's surface. Both LiDAR and SAR offer complementary information for HSI [15], [16]. In order to better use the complementary information from different modalities and to enhance the classification performance, multimodal RS data joint processing has emerged. Following the advanced work in [17] and [18], this study focuses on multimodal RS data classification using HSI, LiDAR, and SAR, which refers to assigning labels to each pixel within the same scene.

To improve the performance of the multimodal RS data classification, feature extraction is a crucial step. The feature extraction methods can use hand-crafted and deep learning

Manuscript received 18 September 2023; revised 9 November 2023; accepted 23 November 2023. Date of publication 27 December 2023; date of current version 3 January 2024. This work was supported in part by the Postdoctoral Researcher Program of China under Grant GZB20230968 and in part by the National Natural Science Foundation of China under Grant 62371169, Grant 61971164, and Grant U20B2041. *(Corresponding author: Yushi Chen.)*

Xin He, Yushi Chen, and Lingbo Huang are with the School of Electronics and Information Engineering, Harbin Institute of Technology (HIT), Harbin 150001, China (e-mail: hexin1@hit.edu.cn; chenyushi@hit.edu.cn; 22B905001@stu.hit.edu.cn).

Danfeng Hong is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: hongdf@aircas.ac.cn).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

Digital Object Identifier 10.1109/TGRS.2023.3344698

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

models, which are represented by morphological operators and CNN-based methods, respectively [19], [20], [21]. Because of the fact that the aforementioned methods may struggle to capture long-range dependencies, transformer-based multimodal RS data methods have attracted much attention in recent years [22]. The attention mechanism in the transformer enables it to attend to different modalities simultaneously and learn their interactions [23]; therefore, transformer-based methods are explored for RS data classification in this research.

In the field of artificial intelligence, foundation models, which are usually based on transformers, have, moreover, provided a glimpse of general artificial intelligence and they have shown their powerfulness in many applications [24]. These models, such as ChatGPT [25], segment anything model (SAM) [26], and DINO [27], are trained on a large-scale dataset with self-learning, which are capable of a wide range of downstream tasks. In general, fine-tuned foundation models have revolutionized various fields by leveraging their immense capacity to generate high-quality outputs and make accurate predictions compared to task-specific models. In the area of computer vision, by using advanced architectures and leveraging vast amounts of data, these foundation models learn visual representations that capture the underlying structure of the data. This allows them to learn representations effectively from extensive data. Consequently, these models have brought about a significant breakthrough by offering powerful backbones for multimodal RS data classification.

How to effectively use foundation models for multimodal RS data classification is, however, still a challenge. The problems include optimizing their architecture, fine-tuning parameters, and selecting appropriate training strategies. Intuitively, to better use these powerful backbones, fine-tuning parameters is simple but useful for multimodal RS data classification; however, the process of fine-tuning often requires a significant amount of memory and storage resources, which is time-consuming and resource-intensive. Fine-tuning is, moreover, vulnerable to catastrophic forgetting as it alters model parameters without considering future domains, and thus, it risks the loss of prior knowledge accumulated during the current adaptation.

To solve the above problems, a foundation model adaptation (FMA) framework is proposed in this article for multimodal RS data classification. Different from fine-tuning, the proposed framework only involves minor modifications to the model architecture without fine-tuning the parameters in the backbone, which makes it suitable for multimodal RS data classification without consuming many resources.

Because of the fact that different modalities exhibit significant differences in terms of data characteristics, integration and analysis of multimodal RS data is a challenging task. The differences among HSI, LiDAR, and SAR datasets present both challenges and opportunities in the joint analysis of multimodal RS data. To address the differences among different RS modalities effectively, a multimodal RS feature alignment method based on the FMA framework is also proposed for classification. The main contributions of this article can be summarized as follows.

1) An FMA framework is proposed for multimodal RS data classification without fine-tuning the parameters. To the best of our knowledge, it is the first attempt to explore the use of a foundation model in the field of multimodal RS data classification.

2) To leverage the potential of foundation models effectively, two learnable modules in the framework are well-designed to facilitate constructive adaptation to multimodal RS data classification, which includes a cross-spatial interaction module and a cross-channel interaction module.

3) To link the features learned in different modalities in the framework, a multimodal RS feature alignment method is proposed for better classification. A coupling score function is designed to explicitly condition each other, which allows modeling the relationships between different modalities, and thus, provides more flexibility to align their representations.

The remainder of this article is organized as follows. Section II gives a comprehensive literature review in the existing researches in the field of multimodal RS data classification. Section III presents detailed information on the proposed methods. Section IV offers a comprehensive analysis of the experimental results. Finally, Section V concludes with a summary.

II. Related Work

A. Classical Multimodal RS Data Classification

The classical multimodal learning methods for RS classification are often related to morphological-based operators and manifold learning [28]. For example, Liao et al. combined spectral information in HSI with information extracted by a morphological operator. Similarly, Khodadadzadeh et al. [29] used different types of morphological operators to fuse HSI and LiDAR data. Based on morphological operators, Rasti et al. [30], furthermore exploited an orthogonal total variation component analysis approach based on the features extracted by extinction profiles, which merged spectral and spatial information from HSI and elevation information from LiDAR. Later, Rasti et al. [31] improved extinction profiles named multiextinction profiles to extract features from HSI and LiDAR, and then a sparse and low-rank component analysis strategy was proposed to fuse their respective features. Another way to learn different modality representations is based on manifold learning. Inspired by the ideas in [32], Hu et al. proposed a manifold feature alignment strategy of different modalities (i.e., HSI and SAR) to accomplish heterogeneous data fusion. Compared to the deep learning-based multimodal RS data classification, the classical methods may have limited feature extraction ability.

B. CNN-Based Multimodal RS Data Classification

In recent years, owing to the powerful feature extraction ability, deep learning-based multimodal methods for RS classification have been developed. According to model architecture, these methods can be roughly divided into two categories: CNN-based and Transformer-based multimodal RS data classification. In terms of CNN-based multimodal RS data classification, Peng et al. [33] offered an overview of domain adaptation methods, encompassing discrepancy-based and adversarial-based adaptations. To effectively exploit the spectral-spatial joint information for domain adaptations of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

HSI classification, Huang et al. [34], furthermore, proposed a two-branch attention adversarial domain adaptation method, which adjusted the distribution between the source and target domains and extract domain-invariant spatial-spectral features. In addition, most CNN-based multimodal RS data classification fused the features from different modalities by concatenation. More specifically, Ghamisi et al. [35] explored an HSI and LiDAR data feature stacking technique by using extinction profiles and CNN. Chen et al. [36] employed a fully connected DNN to fuse the heterogeneous features from MSI and LiDAR data; furthermore, Xu et al. [37] designed a cascade network based on CNN to concatenate features extracted from LiDAR and HSI with a shortcut path. Using CNN to extract the pixel-level features and then concatenating these features for RS classification, other methods pay attention to extracting the semantic information based on the CNN, which was validated on the very high-resolution data, HSI data, and LiDAR data [38]. Although the convolution paradigm has generated excellent results in multimodal RS data classification, it is difficult for the convolution operation to relate spatially distant concepts.

C. Transformer-Based Multimodal RS Data Classification

Transformer outperforms CNN in capturing long-range dependencies and contextual information. Unlike CNN, which primarily concentrates on local features, the transformer can proficiently model feature relationships across the entirety of the RS data, resulting in enhanced classification accuracy. With the characteristics of a self-attention mechanism, the transformer offers a promising new paradigm for multimodal RS data classification. For example, Zhao et al. [22] proposed a fractional Fourier image transformer to effectively extract multimodal heterogeneous features and reduce information loss. For the joint classification of HSI and LiDAR data, Roy et al. [39] deployed a new multimodal fusion transformer (MFT); the LiDAR data are used as an external classification token to the encoder of the transformer, which provides complementary information for HSI. To enhance multimodal information fusion ability, Zhang et al. [40] incorporated the multiscale information into the multimodal fusion process; moreover, in the case of multispectral and panchromatic data integration, to alleviate the redundant information of multimodal data, Ma et al. [41] devised a subspace similar recombination module based on nonnegative matrix factorization and a self-attention mechanism to improve classification performance; nevertheless, the limited ability of these approaches in representing multimodal RS data persists due to the challenge of effectively capturing the complex relationships between different modalities.

D. Foundation Model

A foundation model refers to a kind of deep learning model that serves as a fundamental building block for various downstream tasks [24]. It is designed to learn general features and representations from large-scale datasets, which can then be fine-tuned or extended for specific applications.

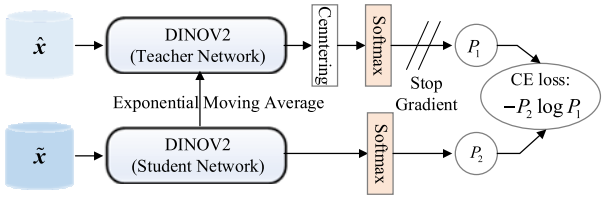


Fig. 1. Diagram of self-distillation.

The development of foundation models has led to significant advancements in deep learning. Researchers continuously explore new architectures, training techniques, and datasets to improve the performance and generalization capabilities of these models. The typical foundation models range from natural language processing (NLP) to computer vision. The typical approaches include bidirectional encoder representation from transformers (BERT) [42], iBOT [43], text-to-text transfer transformer (T5) [44], ChatGPT, general pretrained transformer-3 (GPT-3) [45], CodeGeeX (Codex) [46], DALL-E [47], contrastive language-image pretraining (CLIP) [48], DINO [49], and so on.

Specially, DINOv2, which is created by Meta Research, is a well-known foundation model based on the transformer. Compared to other foundation models (i.e., DINO and iBOT), there are three advantages of DINOv2 [27]. First, DINOv2 is the first self-supervised learning method for training a computer vision model, where a model is trained using images without labels; therefore, DINOv2 saves resources to label data.

Second, DINOv2 uses a larger training dataset, which contains 142 million images covering a number of key visual domains in data collection. By using these images to train the model, DINOv2 provides high-quality visual features that may be used for visual tasks, such as image categorization, instance retrieval, video comprehension, depth estimation, and many more tasks.

Third, multiple improvements to train models on a larger scale are explored in DINOv2, which includes self-distillation, fast and memory-efficient attention, nested tensors in self-attention, efficient stochastic depth, and fully shared data parallel. Specifically, DINOv2 leverages self-distillation techniques to condense a large model named the teacher model into a smaller one named the student model, which exhibits similarities with knowledge distillation. The detailed process is shown in Fig. 1.

A single pair of views $(\hat{\boldsymbol{x}}, \tilde{\boldsymbol{x}})$ of the input $\boldsymbol{x}$ is fed into the teacher network and student network, which have the same architecture and various parameters. The teacher network's outputs are centered by computing the mean over the batch. By using the output features P_1 and P_2 extracted by the class token of ViT, the cross-entropy loss is calculated between the features extracted from the student network and the teacher network. The teacher parameters are updated through the use of an exponential moving average calculated from the student parameters, which makes knowledge distillation particularly useful with DINOv2; moreover, fast and memory-efficient attention, nested tensors in self-attention, efficient stochastic depth, and fully shared data parallel are also explored in DINOv2. The student model, therefore,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

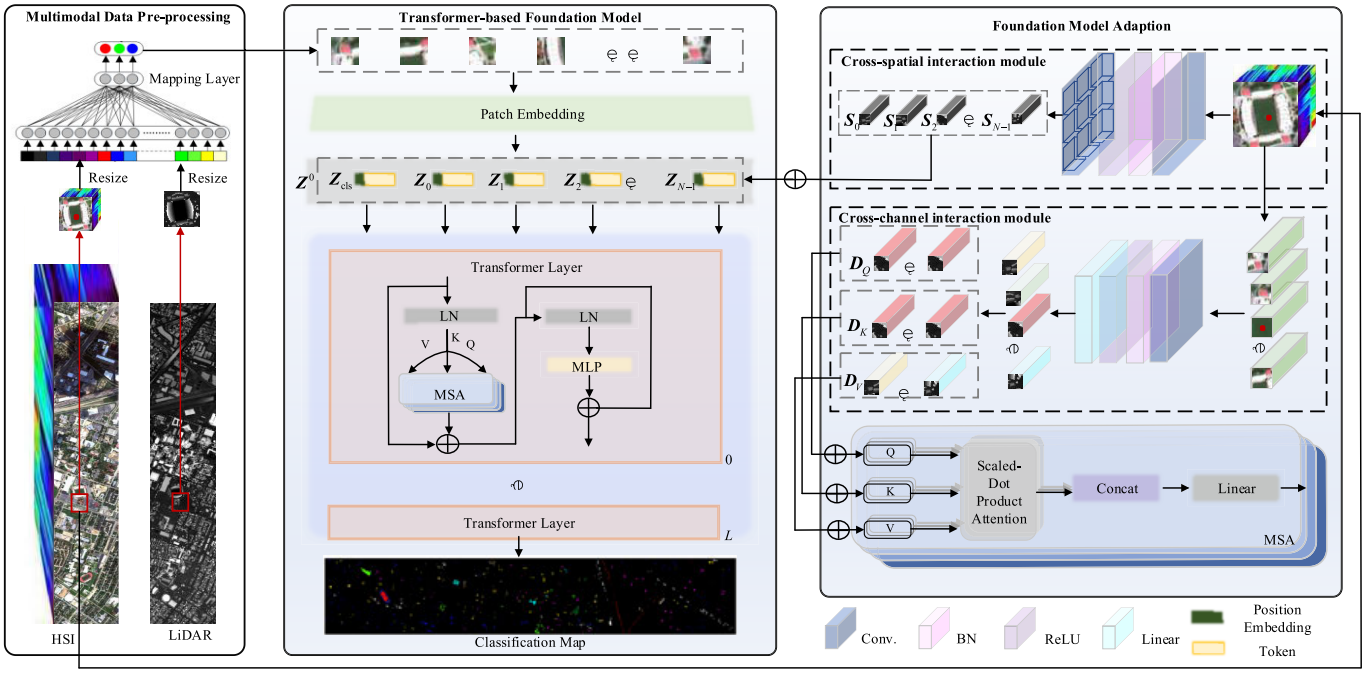


Fig. 2. Overall framework of the proposed FMA for multimodal RS data classification. The proposed FMA framework for multimodal RS data classification with three steps: (a) Multimodal data preprocessing, (b) details of the transformer-based foundation model, and (c) details of the foundation model adaption. The foundation model adaption, moreover, consists of two different learnable modules: cross-spatial interaction module and cross-channel interaction module.

learns class relationships, decision boundaries, and model generalization from the teacher model. DINOv2 is trained to forecast the correlations among various patches within an image, enabling it to gain a comprehensive understanding of structures.

With the characteristics of self-supervised learning, large dataset, and multiple new training strategies, DINOv2 learns better representations and achieves higher performance. Because of the challenges posed by differences between data, applying foundation models to RS classification faces challenges.

III. Method

A. Architecture Overview

Foundation models have been trained on large-scale datasets from diverse domains, which enables them to capture discriminative features from datasets; therefore, these models have powerful generalization capabilities. Foundation models are, moreover, designed with different architectures, which adapt different input modalities and incorporate fusion techniques to effectively combine information from diverse sources; therefore, they are able to learn intricate relationships and spatial dependencies between different image features.

The captured high-level representations allow foundation models to perform well on a wide range of tasks. The training dataset used for foundation models and multimodal remote RS dataset, however, differs significantly, making it challenging to directly apply foundation models for multimodal RS data classification. To address this problem, this study proposes a framework called FMA to more effectively use the parameters for multimodal RS data classification. Fig. 2 depicts the framework of the proposed FMA for multimodal RS data classification.

Specifically, the FMA consists of three steps. The Houston dataset, taken as an example, includes the HSI image and LiDAR data. In the first step, named multimodal data preprocessing, the chosen samples in the HSI image and LiDAR data are fed into the mapping layer to match the dimension of the input of the foundation model. Then, the output concatenated features are fed to the foundation model.

In the second step, which is about the transformer-based foundation model, the outputs of the first step are divided into several patches as the input of the foundation model. Then, these patches are fed into the patch embedding layer to generate a series of tokens. Next, these tokens are the input of transformer layer for feature learning.

In the third step, named foundation model adaption, two well-designed learnable modules, i.e., cross-spatial interaction module and cross-channel interaction module, are attached to the foundation model. Especially to adapt the foundation model for the multimodal RS data classification task, the cross-spatial interaction module generates several tokens along the spatial dimension to extract spatial features, which are attached to the existing tokens in the second step. The cross-channel interaction module extracts the spectral information for the Houston image, which is attached to $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}$ in the multihead self-attention (MSA) of the transformer layer in the second step. Note that the cross-spatial interaction and cross-channel interaction propagate the spatial and spectral information for the detailed multimodal RS classification task on the Houston image. Finally, the class token in transformer backbone is used for classification.

B. Transformer-Based Foundation Model

Compared to other models (e.g., CNN), the transformer is able to model long-range dependencies, which makes it a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

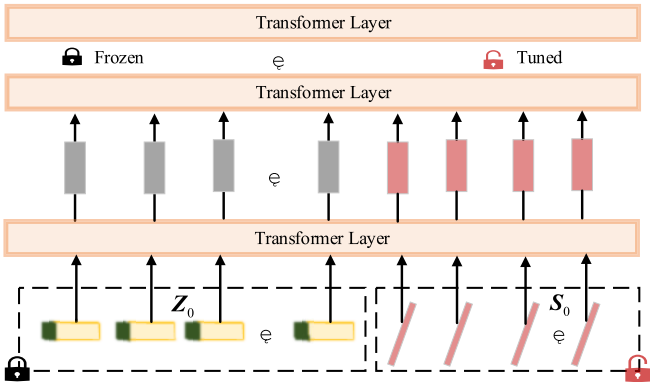


Fig. 3. Cross-spatial interaction module for propagation.

powerful tool for NLP and computer vision tasks. For example, the relationships between words that are far apart in a sentence can be captured by the transformer.

In general, a transformer-based foundation model divides the processed data $\boldsymbol{x}$ into partition patches, whose length is denoted as N. Afterward, the patches are fed into a linear layer, resulting in the formation of a set of embeddings named patch token $\boldsymbol{Z}_n \in \mathbb{R}^d$. To incorporate global information from the entire patches, an additional learnable token named class token $\boldsymbol{Z}_{\text{cls}}$ is added to the patch tokens $\boldsymbol{Z}^0 \in \mathbb{R}^d$, which is represented by:

$$\boldsymbol{Z}^0 = \left[\boldsymbol{Z}_{\text{cls}}, \boldsymbol{Z}_0, \ldots, \boldsymbol{Z}_{\text{N}-1}\right]. \tag{1}$$

Obviously, the core element of the transformer-based foundation model is the self-attention mechanism. There are three vectors in the self-attention mechanism, which maps a query and a set of key-value pairs to an output, and the output is computed by the weighted sum of the values. The three vectors are the query $\boldsymbol{Q}$ with dimension d_k, the key $\boldsymbol{K}$ with dimension d_k, and the value $\boldsymbol{V}$ with dimension d_v. The attention is computed as follows:

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_k}}\right)\boldsymbol{V}. \tag{2}$$

By extracting features by the self-attention mechanism, the patch token representations are updated. Instead of performing a single attention function, it has been proved that projecting the queries, keys, and values f times is effective. The above process is MSA, which is computed as:

$$\text{MSA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{Contact}(\mathbf{head}_1, \ldots, \mathbf{head}_h)\boldsymbol{W}^O \tag{3}$$

$$\mathbf{head}_i = \text{Attention}\left(\boldsymbol{Q}\boldsymbol{W}_i^Q, \boldsymbol{K}\boldsymbol{W}_i^K, \boldsymbol{V}\boldsymbol{W}_i^V\right) \tag{4}$$

where $\boldsymbol{W}_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $\boldsymbol{W}_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $\boldsymbol{W}_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$, $\boldsymbol{W}^O \in \mathbb{R}^{r \times d_v \times d_{\text{model}}}$, and $d_k = d_v = d_{\text{model}}/f$.

In the encoder of ViT with L layers, the MSA is succeeded by an MLP layer with GELU activation, and a residual connection is added after each module. In addition, an LN layer is always behind each MSA module and MLP layer, which is defined as:

$$\boldsymbol{H}^{l-1} = \text{MSA}\left(\text{LN}\left(\boldsymbol{Z}^{l-1}\right)\right) + \boldsymbol{Z}^{l-1}, \quad l = 0, \ldots, L-1 \tag{5}$$

$$\boldsymbol{Z}^{l} = \text{MLP}\left(\text{LN}\left(\boldsymbol{H}^{l-1}\right)\right) + \boldsymbol{H}^{l-1}, \quad l = 0, \ldots, L-1. \tag{6}$$

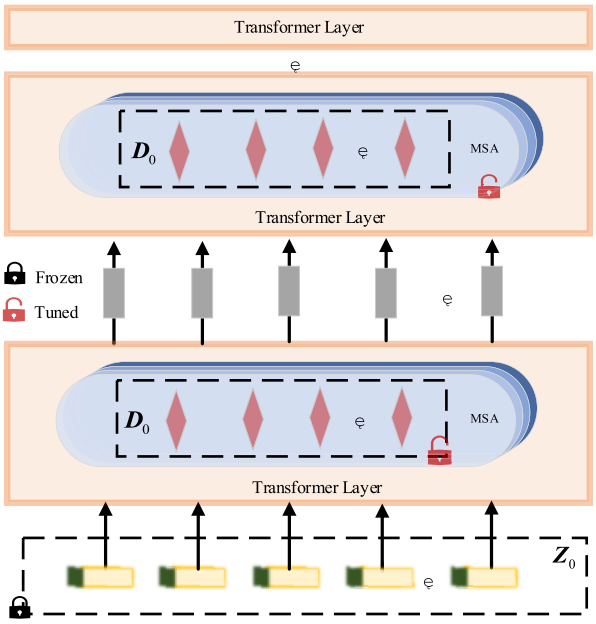


Fig. 4. Cross-channel interaction module for propagation.

Among the existing foundation models, DINOv2 learns information from unlabeled data, which has been found to perform effectively on many image analysis tasks. In this study, we use it to provide a diverse range of pretrained visual models that have been trained with various ViT architectures, such as ViT-Small, ViT-Base, and ViT-Large models. The pretrained model exhibits exceptional performance without fine-tuning and makes it an ideal choice as a backbone for various downstream tasks. If we can compute embeddings based on these backbones for multimodal RS data classification with the unique approach to learning visual representations from large datasets, the classification performance would be enhanced. To achieve this goal, there are two main issues that need to be considered, one is the difference between the training datasets of the foundation model and multimodal RS dataset, and the other is feature alignment between various RS modalities.

C. Foundation Model Adaptation for Multimodal RS Data Classification

The foundation models based on the ViT, such as DINOv2, produce high-performance visual features and provide several pretrained visual models; however, effectively harnessing these pretrained visual models poses challenges, particularly in multimodal RS data classification. Typically, the finetuning strategy is employed for model adaptation. The acquisition of training samples is, however, costly and time-consuming. Fine-tuning all model parameters of a foundation model, moreover, may neglect the learned visual features during the pretraining process. In this case, the overfitting problem occurs under limited multimodal RS training samples, which results in poor generalization.

In multimodal RS data, there always exist several unimodalities. These modalities can vary in their characteristics and organization, but one common aspect is the presence of spatial dimension and channel dimension. Spatial dimension refers to the spatial or positional information contained within

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

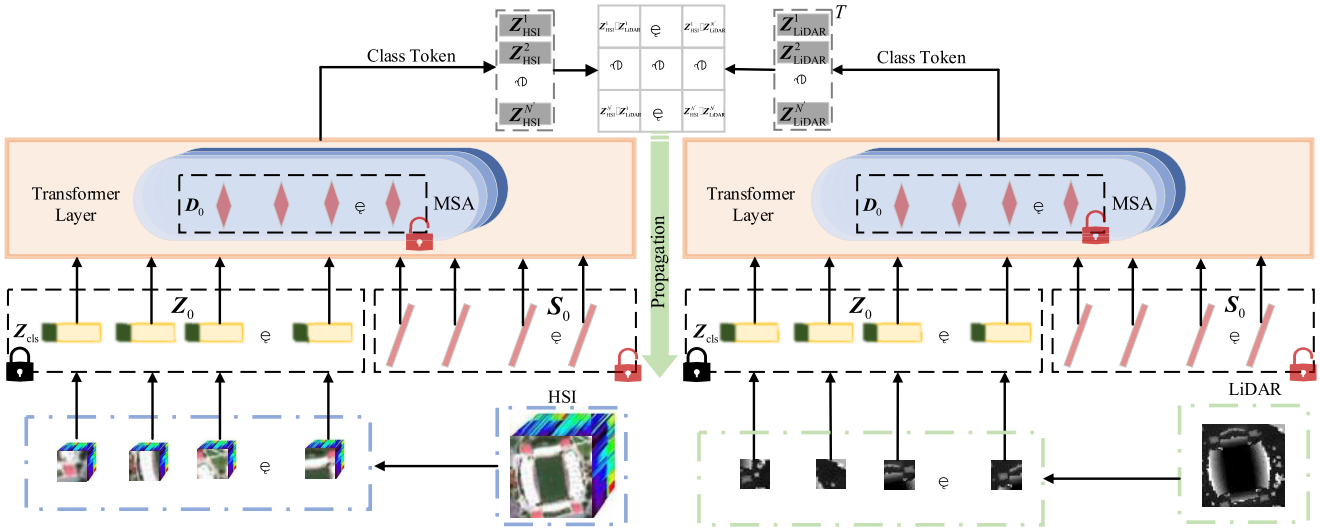


Fig. 5. Framework of the proposed FMA^2 for multimodal RS data classification.

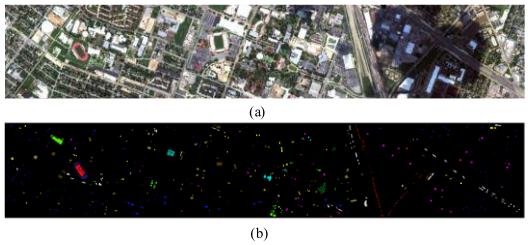


Fig. 6. (a) False color composite image and (b) ground reference map on the Houston dataset.

the dataset. In RS data, spatial dimension includes spatial information in an HSI dataset, height in a LiDAR dataset, and backscattering of targets in an SAR dataset. On the other hand, channel dimension refers to different channels. The channel dimension can be one, two, or several hundred. For example, in HSI, there are hundreds of channels that represent spectral information, while in polarized SAR, the channel information indicates the polarization information.

To overcome the bottleneck in a limited amount of data, a FMA method named FMA is proposed. FMA designs a cross-spatial interaction module and cross-channel interaction module to efficiently adapt visual features in the foundation model. Taking the Houston image as an example, to build a unified FMA framework for multimodal RS data classification, in the cross-spatial interaction module, a learnable patch token set $\boldsymbol{S}^0$ is generated along the spatial dimension of HSI, which is attached to the fixed patch token set $\boldsymbol{Z}^0$.

Specifically, the fixed patch token set $\boldsymbol{Z}^0$ fuses the information from the HSI and LiDAR datasets. To match the dimensions of the input of foundation models, the third dimension of different datasets is mapped into three to feed into the backbone of foundation models; then, the mapped datasets are concatenated to generate $\boldsymbol{Z}^0$ in the subsequent steps. In terms of the learnable patch token set $\boldsymbol{S}^0$ in the cross-spatial interaction module, it is built to model the token relationships between all multimodalities and spatial information in HSI, as displayed in Fig. 3.

The learnable patch token sets $\boldsymbol{S}^0$ in the cross-spatial interaction module are generated by a CNN, as shown at the right part in Fig. 2. The CNN includes a 1×1 convolution layer, a BN layer, a ReLU activation function, and a 3×3 convolution layer. Suppose the chosen HSI sample is represented by $\boldsymbol{x}_{\mathrm{HSI}} \in \mathbb{R}^{H \times W \times C}$, where H and W indicate the two spatial dimensions, and C represents the channel dimension. After extracting the spatial features in HSI, the output features $\boldsymbol{S}$ of CNN is described by

$$\boldsymbol{S} = \mathrm{CNN}(\boldsymbol{x}_{\mathrm{HSI}}) \tag{7}$$

where $\boldsymbol{S} \in \mathbb{R}^{H \times W \times C'}$, and $\mathrm{CNN}(\cdot)$ represents the entire CNN process of extracting features along the spatial dimension.

Then $\boldsymbol{S}$ are partitioned into nonoverlapped patches, which generates the $\boldsymbol{S}^0$, similarly in (1)

$$\boldsymbol{S}^0 = \left[\boldsymbol{S}_0, \ldots, \boldsymbol{S}_{\mathrm{N}-1}\right] \tag{8}$$

whose dimension is the same as the tokens in the Transformer-based foundation model.

Based on the (5) and (6), $\boldsymbol{S}^0$ is concatenated with the $\boldsymbol{Z}^0$, which are represented as:

$$\tilde{\boldsymbol{Z}}^0 = [\boldsymbol{Z}^0 || \boldsymbol{S}^0] \tag{9}$$

$$\tilde{\boldsymbol{H}}^{l-1} = \mathrm{MSA}\left(\mathrm{LN}\left(\tilde{\boldsymbol{Z}}^{l-1}\right)\right) + \tilde{\boldsymbol{Z}}^{l-1}, \quad l = 0, \ldots, L-1 \tag{10}$$

$$\tilde{\boldsymbol{Z}}^{l} = \mathrm{MLP}\left(\mathrm{LN}\left(\tilde{\boldsymbol{H}}^{l-1}\right)\right) + \tilde{\boldsymbol{H}}^{l-1}, \quad l = 0, \ldots, L-1. \tag{11}$$

The token relationships between all multimodalities and spatial information are, therefore, modeled by interacting with the fixed patch tokens in the self-attention layers for multimodal RS data classification.

To increase the token relationships between all multimodalities along the channel dimension, furthermore, the cross-channel interaction module is built, which is shown in Fig. 4.

During layer-wise computations of the transformer-based foundation model, such as the query-key-value computations, the cross-channel interaction module is explored to generate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

TABLE I
LABEL INFORMATION OF THE HOUSTON DATASET

No.	Color	Name	Numbers
1		Healthy Grass	1251
2		Stressed Grass	1254
3		Synthetic Grass	697
4		Tree	1244
5		Soil	1242
6		Water	325
7		Residential	1268
8		Commercial	1244
9		Road	1252
10		Highway	1227
11		Railway	1235
12		Parking Lot1	1233
13		Parking Lot2	469
14		Tennis Court	428
15		Running Track	660
Total			15029

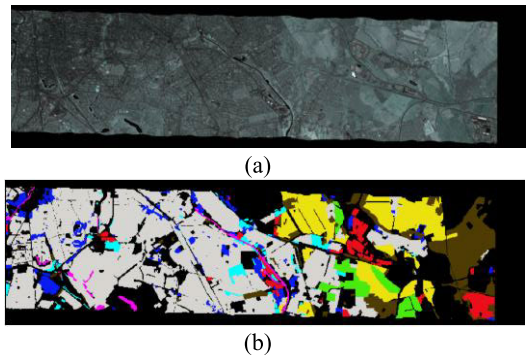


Fig. 7. (a) False color composite image and (b) ground reference map on the Berlin dataset.

TABLE II
LABEL INFORMATION OF THE BERLIN DATASET

No.	Color	Name	Numbers
1		Forest	51954
2		Residential Area	268642
3		Industrial Area	19566
4		Low Plants	59282
5		Soil	17426
6		Allotment	13305
7		Commercial Area	24824
8		Water	6672
Total			464671

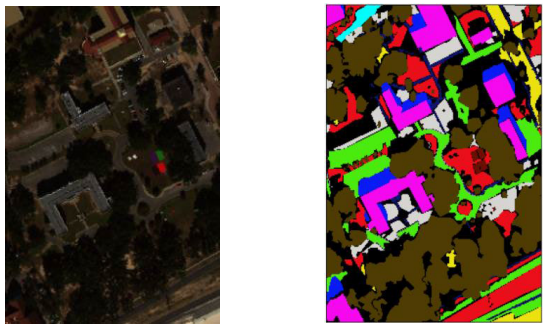


Fig. 8. (a) False color composite image and (b) ground reference map on the MUFFL dataset.

TABLE III
LABEL INFORMATION OF THE MUUFL DATASET

No.	Color	Name	Numbers
1		Trees	23246
2		Mostly grass	4270
3		Mixed ground surface	6882
4		Dirt and sand	1826
5		Road	6687
6		Water	466
7		Building shadow	2233
8		Building	6240
9		Sidewalk	1385
10		Yellow curb	183
11		Cloth panels	269
Total			53687

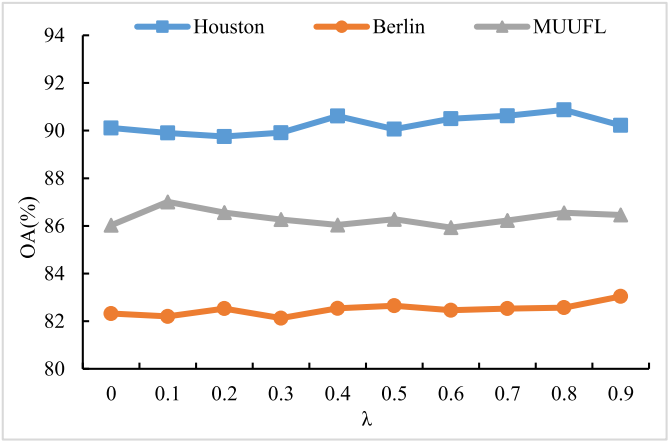


Fig. 9. Relationship between the values of λ and OA.

spectral features, which are attached to the MSA without significantly increasing computational cost.

Specifically, the cross-channel interaction module takes the mean spectral features of each patch of $\boldsymbol{x}_{\mathrm{HSI}}$ as input. Then, these features are fed to a 1×1 convolution layer, a BN layer, a ReLU activation function, and an MLP layer. Suppose the output is $\boldsymbol{D}^{'} = [\boldsymbol{D}_0, \ldots, \boldsymbol{D}_{\mathrm{N}'}]$, whose dimension is the same as the transformer layer in the transformer-based foundation model. Next, $\boldsymbol{D}^{'}$ is used to produce $\boldsymbol{D}_Q$, $\boldsymbol{D}_K$, and $\boldsymbol{D}_V$, which are added to the MSA in the foundation module. $\boldsymbol{D}_Q$ and $\boldsymbol{D}_K$ are calculated from the middle of $\boldsymbol{D}^{'}$, which is $\boldsymbol{D}_{(\mathrm{N}'+1)/2}$. Then, the dimension of $\boldsymbol{D}_Q$ and $\boldsymbol{D}_K$ are repeated for N' times to guide the self-attention mechanism in FMA to capture the information around the center pixel of $\boldsymbol{x}_{\mathrm{HSI}}$, which is helpful for RS data classification. Here, $\boldsymbol{D}_V$ is equal to $\boldsymbol{D}^{'}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

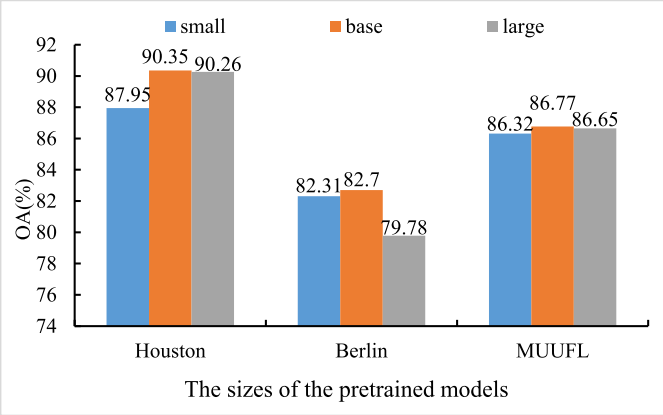


Fig. 10. Relationship between the sizes of the pretrained model and OA.

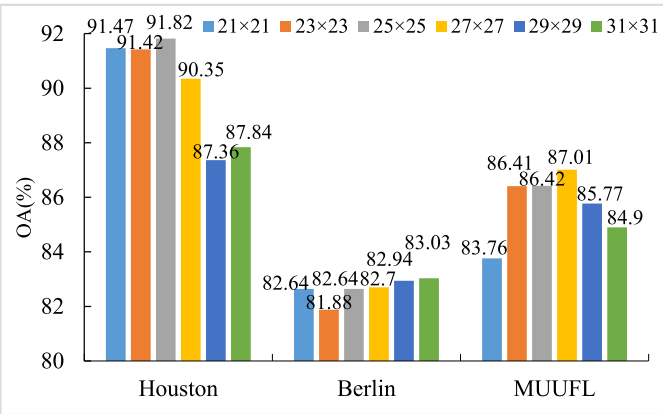


Fig. 11. Results of the proposed FMA with different input data sizes.

Based on the (3) and (4), the computation of MSA is changed into follows:

$$\boldsymbol{g}_i = \text{Attention}\big(\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i\big), \quad i = 1, \ldots, h \tag{12}$$

$$\boldsymbol{Z}' = \text{LN}(\boldsymbol{Z}^0) \tag{13}$$

$$\boldsymbol{Q}_i = \boldsymbol{Z}'\boldsymbol{W}_i^Q, \ \boldsymbol{K}_i = \boldsymbol{Z}'\boldsymbol{W}_i^K, \ \boldsymbol{V}_i = \boldsymbol{Z}'\boldsymbol{W}_i^V \tag{14}$$

$$\tilde{\boldsymbol{Q}}_i = \boldsymbol{Q}_i + \boldsymbol{D}_Q, \ \tilde{\boldsymbol{K}}_i = \boldsymbol{K}_i + \boldsymbol{D}_K, \ \tilde{\boldsymbol{V}}_i = \boldsymbol{V}_i + \boldsymbol{D}_V \tag{15}$$

where $\boldsymbol{D}_Q$, $\boldsymbol{D}_K$, and $\boldsymbol{D}_V$ indicate the spectral features extracted by the cross-channel interaction module, being added to $\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i$ of the ith head, respectively.

Thus, FMA fuses the relationships between all multimodalities, spatial features, and spectral features by only adjusting the cross-spatial interaction module and the cross-channel interaction module while the weights of the backbone of foundation models are frozen.

D. Multimodal RS Feature Alignment for Classification

Joint exploitation of multimodal RS data is capable of providing complementary information; however, the disparities between different modalities are large. These differences arise from inherent characteristics and representations of each modality.

To close the gaps between different modalities, FMA with alignment (FMA^2) is proposed based on the FMA for multimodal RS data classification. Fig. 5 shows its framework, which consists of an HSI module and a LiDAR module.

Specifically, $\boldsymbol{Z}^0$ is replaced with single-modality RS data, such as $\boldsymbol{Z}^{\text{HSI}}$, $\boldsymbol{Z}^{\text{LiDAR}}$, or $\boldsymbol{Z}^{\text{SAR}}$. $\boldsymbol{S}^0$ and $\boldsymbol{D}^0$ in FMA^2 are unchanged because in FMA, the foundation model possesses significant advantages in feature extraction ability, enabling it to capture general features effectively from extensive data. To better use these powerful general features, $\boldsymbol{Z}^0$ is from the foundation model; thus, $\boldsymbol{Z}^0$ is used to capture the general spatial features from the foundation model. While $\boldsymbol{S}^0$ and $\boldsymbol{D}^0$ are designed to extract multimodal-specific representations. With different input single modalities, the general spatial features also exhibit differences since the spatial resolutions of HSI and LiDAR are different. FMA^2 aims to further tackle the disparities among these general features rather than using the multimodal-specific representations (i.e., spectral information and height information). $\boldsymbol{S}^0$ and $\boldsymbol{D}^0$ are, therefore, the same as FMA, which are from HSIs.

Each modality is fed into the FMA, to promote coupling between the HSI image and LiDAR or SAR data, a coupling score function is designed for multimodal RS data.

After feature extraction by the transformer layer, the class token $\boldsymbol{Z}_{\text{cls}}$ is always the presentation vector. In the Houston RS data, the output class token of the transformer layer on the modalities HSI and LiDAR are denoted as $\boldsymbol{Z}_{\text{HSI}}^{\text{cls}} = [\boldsymbol{Z}_{\text{HSI}}^1, \boldsymbol{Z}_{\text{HSI}}^2, \ldots, \boldsymbol{Z}_{\text{HSI}}^{N'}]$ and $\boldsymbol{Z}_{\text{LiDAR}}^{\text{cls}} = [\boldsymbol{Z}_{\text{LiDAR}}^1, \boldsymbol{Z}_{\text{LiDAR}}^2, \ldots, \boldsymbol{Z}_{\text{LiDAR}}^{N'}]$, respectively. Here, N' denotes the dimension of the class token. Then, $\boldsymbol{Z}_{\text{HSI}}^{\text{cls}}$ and $\boldsymbol{Z}_{\text{LiDAR}}^{\text{cls}}$ are normalized by column, before being fed to the coupling score function

$$\mathcal{L}_c = \sum_{i=1}^{N'}\sum_{j=1}^{N'}\left(\boldsymbol{Z}_{\text{HSI}}^i \cdot \boldsymbol{Z}_{\text{LiDAR}}^j\right)_{i\neq j} - \sum_{i=1}^{N'}\left(\boldsymbol{Z}_{\text{HSI}}^i \cdot \boldsymbol{Z}_{\text{LiDAR}}^j\right). \tag{16}$$

Finally, based on the FMA, the optimization loss in FMA^2 is defined by

$$\mathcal{L} = \underset{\{\boldsymbol{S}^0\},\{\boldsymbol{D}^0\}}{\text{argmin}}\, L_{\text{CE}}(\text{FC}(\boldsymbol{x}), y) + \lambda\mathcal{L}_c \tag{17}$$

where $\mathcal{L}_{\text{CE}}$ is cross entropy loss, $\boldsymbol{x}$ is the output features of FMA of the input data, and y is the corresponding label. Here, FC represents the fully connected layer, which maps the output features into the number of classes, and λ is a hyperparameter, which balances the loss between $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_c$.

By optimizing the (17), FMA^2 promotes the coupling between multimodal data to further enhance the joint classification performance.

IV. Experiments

In this section, we conduct a comprehensive evaluation of the performance of the proposed methods on three widely used benchmark multimodal RS datasets.

A. Data Description

In the experiments, three multimodal RS datasets, including Houston, Berlin, and MUUFL, from various types of sensors

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION STUDY ON THE HOUSTON DATASET WITH MULTIMODALITIES

Z_0	S_0	D_0	OA	AA	K
√	√	√	90.35 ± 0.70	90.11 ± 0.88	89.57 ± 0.75
√	√		90.10 ± 1.19	89.92 ± 1.47	89.29 ± 1.28
√		√	85.74 ± 0.89	84.69 ± 1.00	84.58 ± 0.97
	√	√	88.79 ± 1.07	89.01 ± 1.15	87.88 ± 1.16
√			85.17 ± 1.52	85.17 ± 1.98	83.97 ±1.65
	√		83.47 ± 1.17	84.11 ± 1.04	82.13 ± 1.26
		√	85.24 ± 1.68	85.61 ± 1.50	84.05 ± 1.81

TABLE V
ABLATION STUDY ON THE BERLIN DATASET WITH MULTIMODALITIES

Z_0	S_0	D_0	OA	AA	K
√	√	√	82.70 ± 0.41	59.68 ± 2.03	71.38 ± 0.67
√	√		81.90 ± 0.77	55.82 ± 1.67	69.74 ± 1.30
√		√	81.08 ± 1.25	53.64 ± 1.28	68.12 ± 1.72
	√	√	82.59 ± 0.65	58.65 ± 1.62	71.04 ± 1.26
√			74.50 ± 3.14	36.20 ± 5.10	53.62 ± 6.30
	√		81.93 ± 0.73	56.14 ± 2.15	69.92 ± 1.60
		√	81.85 ± 0.57	58.31 ± 0.23	70.58 ± 0.55

TABLE VI
ABLATION STUDY ON THE MUUFL DATASET WITH MULTIMODALITIES

Z_0	S_0	D_0	OA	AA	K
√	√	√	86.77 ± 0.84	57.56 ± 1.94	82.16 ± 1.15
√	√		76.26 ± 1.11	44.84 ± 1.96	67.76 ± 1.50
√		√	83.11 ± 1.49	51.47 ± 2.79	77.15 ± 2.02
	√	√	86.39 ± 1.32	57.33 ± 1.48	81.65 ± 1.84
√			76.25 ± 1.17	44.32 ± 2.84	67.72 ± 1.80
	√		74.12 ± 1.08	42.90 ± 1.38	64.93 ± 1.24
		√	85.77 ± 0.37	60.29 ± 2.12	81.00 ± 0.48

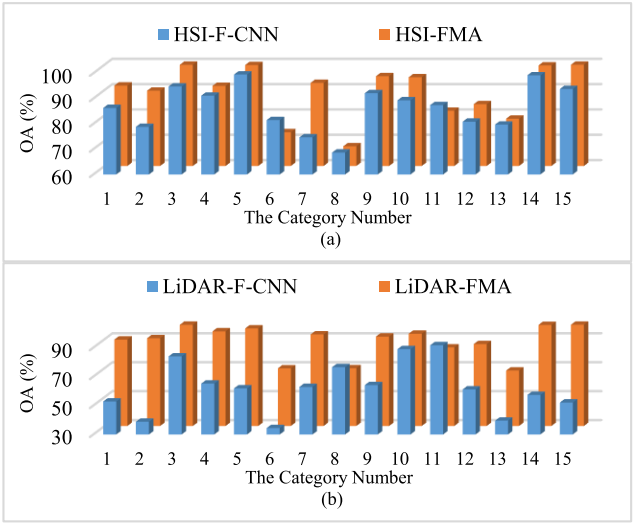


Fig. 12. Performance of different methods with different modalities on the Houston dataset. (a) OA of each class of F-CNN or FMA with HSI. (b) OA of each class of F-CNN or FMA with LiDAR.

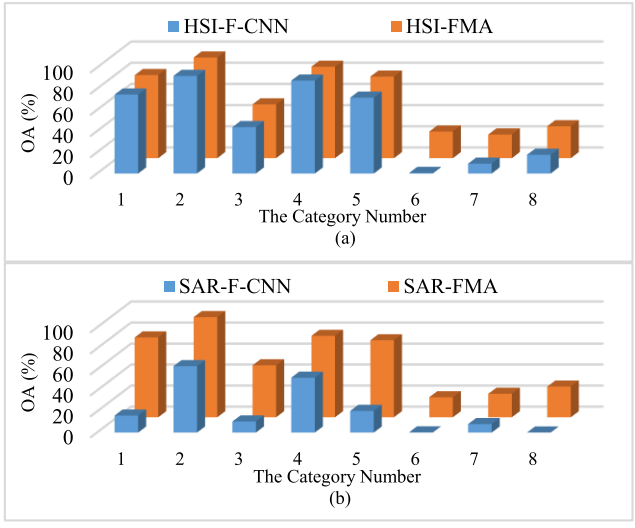


Fig. 13. Performance of different methods with different modalities on the Berlin dataset. (a) OA of each class of F-CNN or FMA with HSI. (b) OA of each class of F-CNN or FMA with SAR.

(i.e., the HSI, SAR, and LiDAR) are considered to validate the performance of the proposed approaches.

The Houston dataset was captured by the Compact Airborne Spectrographic Imager (CASI) sensor over the campus of University of Houston, which was one of the datasets in 2013 Data Fusion Contest organized by the IEEE Geoscience and Remote Sensing Society. The spatial size of the image is 340 × 1905 with the wavelength ranging from 364 to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

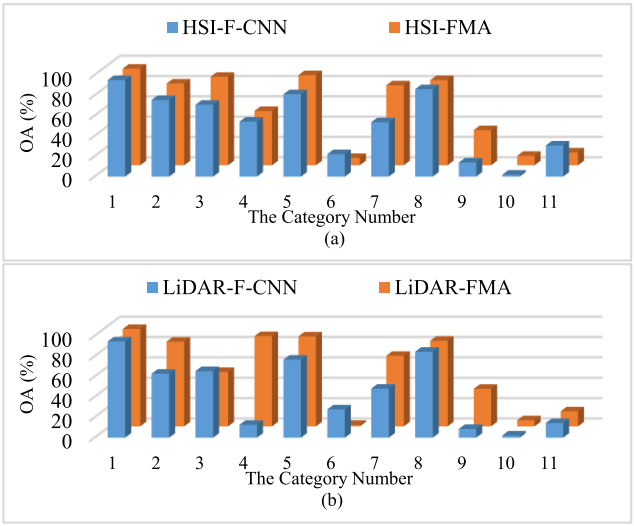


Fig. 14. Performance of different methods with different modalities on the MUUFL dataset. (a) OA of each class of F-CNN or FMA with HSI. (b) OA of each class of F-CNN or FMA with LiDAR.

1046 nm. There are two sources included in this dataset, named the HSI image and LiDAR data, where the HSI image comprises 144 bands and LiDAR data consists of one band. The false color composite and ground reference map are displayed in Fig. 6 and the label information about 15 land cover classes are reported in Table I.

The Berlin dataset provides a comprehensive representation of the Berlin urban area at a GSD of 30 m. This scene includes 797 $\times$ 220 pixels with 244 spectral bands ranging from 0.4 to 2.5 μm. Further information about this dataset is available in [50]. To match the image size of the SAR image, which consists of 1723 $\times$ 476 pixels, the nearest neighbor interpolation technique is used on the HSI image. The false color composite and ground reference map are shown in Fig. 7, and the label information about eight categories is reported in Table II.

The MUUFL dataset was acquired using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in November 2010 over the campus of the University of Southern Mississippi. The HSI of this dataset contains 325 $\times$ 220 pixels with 72 spectral bands ranging from 375 to 1050 nm. After removing water absorption bands the HSI image is left with 64 bands. Additionally, the Gemini ALTM LiDAR sensor is used to capture the LiDAR data. The false color composite and ground reference map are shown in Fig. 8, and the label information about 11 categories is reported in Table III.

B. Implementation Details

1) Algorithm Configuration: The experiments were conducted on a computer equipped with an Intel Core i7-11700 processor clocked at 2.5 GHz, 32 GB of DDR4 RAM with dual data rate, and an NVIDIA GeForce RTX 3070 Ti GPU. The codes are implemented in Python version 3.9 and Pytorch version 1.1.0. The model was trained using the Adam optimizer with an initial learning rate of 5 $\times$ 10^{-5}, batch size of 8, and 150 epochs.

To quantitatively evaluate the performance, three classic metrics were used, namely overall accuracy (OA), average accuracy (AA), and kappa coefficient (K); furthermore, we generated classification maps using all methods and visually compared them for qualitative assessment.

2) Comparison Methods: Several state-of-the-art classification methods, including extended morphological profiles-support vector machine (EMP-SVM) [51], transfer learning based on CNN (HT-CNN) [52], spectral-spatial feature tokenization transformer (SSFTT) [53], fusion CNN (F-CNN) [36], fusion transformer (MFT) [38], and cross-channel reconstruction module (CCRNet) [54] are selected for comparison with our proposed FMA and FMA^2.

Specifically, EMP-SVM extracts spatial information from the first three principal components of HSIs. A disk-shaped structuring element is adopted with a size range of 2–8, resulting in 27 spatial features. In order to determine the optimal values for parameters, such as c and γ, a search method is, moreover, conducted in the range of (10^{-3}, 10^{-2}, 10^{-1}, 1, 10^{1}, 10^{2}, and 10^{3}).

HT-CNN combines transfer learning and deep CNN, and the first several layers of VGGNet-16 pre-trained on the ImageNet are transferred to the target CNN model.

SSFTT combines a backbone CNN with a transformer organically. The convolution layers effectively capture low-level convolution spectral-spatial features, which are then converted into semantic tokens.

F-CNN extracts the features of HSI and LiDAR or SAR by using CNN separately; then, these features are directly stacked for multimodal RS classification.

The MFT network is a recently developed transformer-based multimodal fusion model that can classify HSI along with other types of multimodal data. By using other multimodal data as external classification tokens to the transformer encoder, the MFT network can achieve improved generalization performance.

CCRNet incorporates a CCRNet that enables the model to learn more compact fusion representations of various RS data sources. By employing a reconstruction strategy across modalities, CCRNet facilitates more efficient information exchange between different sources.

C. Parameter Analysis

This section delves into an in-depth analysis of various key parameters, including the weight of the coupling score function of FMA^2, the pretrained models of DINOv2, and the size of input data. Specifically, the weight of the coupling score function λ adjusts the degree alignment of the different multimodal RS images. The size of the pretrained models influences the complexity of the model.

1) *The weight of the coupling score function*: To validate the relationship between the values of λ and overall classification, we use the grid search method to find the optimal value of λ on the three multimodal datasets, and the corresponding classification results are displayed in Fig. 9. From the results, it can be seen that OA fluctuates when the value of λ increases. In most cases, compared to the OA when the value of λ is set to 0, OA is generally higher when λ is greater than 0 in most cases. To achieve better performance, we set the default value of λ to 0.4 on the Houston dataset, 0.9 on the Berlin dataset, and 0.1 on the MUUFL dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
CLASSIFICATION RESULTS (VALUES ± STANDARD) OF DIFFERENT METHODS ON THE HOUSTON DATASET

Methods	EMP-SVM	HT-CNN	SSFTT	F-CNN	MFT	CCRNet	**FMA**	**FMA^2**
OA	74.92±3.53	80.20 ± 1.29	87.04 ± 0.57	87.69 ± 2.43	88.62±0.54	81.54± 0.53	90.35 ± 0.70	90.87 ± 1.18
AA	72.92±4.88	79.38 ± 1.68	87.38 ± 0.94	85.66 ± 2.58	87.24±1.07	76.19±0.76	90.11 ± 0.88	90.70 ± 1.68
K	72.84±3.85	78.59 ± 1.40	85.98 ± 0.62	86.68 ± 2.628	87.69±0.59	80.00±0.58	89.57 ± 0.75	90.14 ± 1.28
1	84.66±4.28	89.98±4.34	82.14±3.94	92.92±4.84	91.70±4.63	86.27±0.54	90.25±3.49	90.17±4.06
2	62.37±11.84	77.71±8.25	91.21±5.10	79.51±10.63	89.18±4.70	94.88±0.68	92.61±4.23	94.65±3.68
3	90.30±9.27	96.00±1.72	96.56±1.91	93.18±5.45	98.99±0.61	93.66±2.44	99.88±0.15	99.97±0.06
4	73.85±9.76	72.21±5.37	91.41±3.32	91.20±5.72	95.52±1.95	91.79±0.84	95.30±2.42	94.19±3.10
5	89.22±5.30	95.44±2.80	99.01±1.40	98.06±1.63	97.35±2.11	95.04±0.48	99.70±0.41	99.55±0.55
6	39.77±14.70	66.34±16.29	83.18±2.41	52.05±24.24	60.68±6.45	0.99±0.50	75.30±3.37	79.82±4.13
7	76.06±9.12	75.63±9.32	84.93±5.21	86.63±8.99	88.79±2.13	83.72±1.10	92.82±4.08	93.27±4.10
8	57.59±6.66	63.99±10.91	74.17±6.06	83.34±9.92	91.94±2.70	81.97 ± 0.03	68.28±8.95	73.59±8.88
9	80.99±7.15	80.44±9.28	60.32±16.49	87.49±3.32	72.39±1.55	61.27± 2.63	93.92±1.17	92.37±2.88
10	77.89±11.64	87.48±9.91	91.44±9.52	91.15±4.93	87.93±3.53	90.76± 4.35	95.53±5.33	94.55±4.01
11	83.16±6.45	82.14±13.58	93.07±2.90	88.88±9.21	85.76±1.59	82.62± 0.58	83.70±6.02	84.08±5.92
12	77.48±12.81	81.83±3.34	93.98±4.86	86.64±16.81	86.77±4.08	73.82± 0.79	86.99±7.24	87.99±6.75
13	70.67±31.30	54.68±21.95	69.85±11.21	55.61±33.79	65.60±10.72	35.79 ±5.05	77.36±17.28	76.33±20.81
14	64.02±22.05	93.36±5.83	99.84±0.23	99.81±0.39	96.44±2.52	74.93±10.81	99.95±0.10	99.95±0.10
15	65.74±20.47	73.52±16.44	99.58±0.60	98.42±1.01	99.59±0.58	95.35±3.43	100.00±0.00	99.97±0.06

TABLE VIII
CLASSIFICATION RESULTS (VALUES ± STANDARD) OF DIFFERENT METHODS ON THE BERLIN DATASET

Methods	EMP-SVM	HT-CNN	SSFTT	F-CNN	MFT	CCRNet	**FMA**	**FMA^2**
OA	78.09±0.65	79.39 ± 0.56	80.56 ± 0.48	80.62 ± 1.09	81.78±0.16	77.61± 0.97	82.70 ± 0.41	83.04 ± 0.47
AA	47.78±1.79	48.33 ± 1.51	56.71 ± 0.10	49.92 ± 3.53	61.81±2.98	50.74± 2.88	59.68 ± 2.03	58.59 ± 2.53
K	64.23±2.34	65.01 ± 1.13	68.41 ± 0.09	66.46 ± 1.39	69.39±0.35	62.03± 1.81	71.38± 0.67	71.55± 1.08
1	68.52± 3.93	69.64±5.10	87.25±1.32	80.11±7.41	73.42±1.63	78.40±2.40	79.49±4.22	78.76±5.12
2	96.56± 1.70	94.54±0.88	90.93±2.31	95.77±4.85	94.49±0.21	91.93±0.44	94.34±1.92	95.56±1.45
3	34.51±10.74	41.99±23.27	33.63±8.93	34.04±18.17	56.56±3.58	14.56±0.88	51.02±24.73	51.00±26.40
4	78.61± 2.38	85.32±3.60	82.88±2.62	81.83±6.07	82.22±3.31	75.73±1.25	86.04±4.55	86.27±3.27
5	68.60± 8.10	79.05±9.78	72.92±5.62	70.79±15.33	80.83±4.89	80.66±7.16	82.87±7.70	78.23±10.46
6	3.4±4.88	0.76±1.15	30.70±16.23	7.69±6.74	29.22±16.45	7.86±2.86	27.17±16.30	22.74±15.15
7	28.10± 1.60	11.84±8.74	30.58±1.61	11.54±9.27	14.03±1.62	19.05±2.07	20.43±14.97	20.47±12.99
8	3.92 ±5.55	3.50±5.18	24.79±23.57	17.61±13.39	63.67±8.08	37.76±10.24	36.09±12.53	35.68±15.33

2) *The size of the pretrained models:* There are three types pretrained models of DINOv2, which include a small pretrained model (small), a base pretrained model (base), and a large pretrained model (large). To conduct a comprehensive study of the foundation models on the three datasets, the overall classification results of the aforementioned three sizes of DINOv2 are validated in Fig. 10. It can be seen that in comparison with the small and large pretrained model, the based pretrained model achieves better performance on the all multimodal RS images, and especially, the base pretrained model obtains an overall gain from 87.95% to 90.35% on the Houston dataset compared to the small pretrained model; therefore, the base pretrained model is chosen for the three multimodal RS images.

3) *The size of input data:* To report the influences of the proposed FMA with different sizes of input data (i.e., 23 × 23, 25 × 25, 27 × 27, 29 × 29, and 31 × 31), Fig. 11 shows the corresponding classification results on the three multimodal datasets. It can be seen that FMA obtains the highest OA concerning the input data size of 25 × 25 on the Houston dataset, input data size of 29 × 29 on the Berlin dataset, and input data size of 27 × 27 on the MUUFL dataset. In the experiments, to give a fair comparison with other methods, the input data size is set to 27 × 27 for all the experiments.

D. Ablation Study

1) Each Component of the Proposed FMA: In the proposed FMA, there are three components named $\boldsymbol{Z}_0$, $\boldsymbol{S}_0$, and $\boldsymbol{D}_0$. Among them, $\boldsymbol{Z}_0$ indicates the two multimodalities that include HSI and LiDAR or SAR, $\boldsymbol{S}_0$ represents the spatial information of HSI, and $\boldsymbol{D}_0$ indicates the spectral information of HSI. To access the importance and impact of each component on the model's performance, we remove each from the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
CLASSIFICATION RESULTS (VALUES ± STANDARD) OF DIFFERENT METHODS ON THE MUUFL DATASET

Methods	EMP-SVM	HT-CNN	SSFTT	F-CNN	MFT	CCRNet	**FMA**	**FMA^2**
OA	72.91 ±2.16	79.08 ± 0.61	82.56 ± 1.07	83.01 ± 0.80	83.35 ± 1.36	81.00 ± 0.33	86.77 ± 0.84	87.01 ± 0.19
AA	36.75 ±3.87	50.94 ± 3.25	58.69 ± 3.97	50.13 ± 1.80	52.48 ± 5.47	56.20 ± 2.19	57.56 ± 1.94	58.54 ± 2.44
K	62.05 ±3.79	71.33 ± 0.88	76.57 ± 1.50	76.75± 1.23	77.57± 1.89	74.41 ± 0.48	82.16 ± 1.15	82.51 ± 0.25
1	94.31 ± 1.60	94.71±1.03	94.65±0.41	95.79±1.08	95.91±1.66	95.35±0.05	95.86±0.95	95.74±0.60
2	47.21±12.34	71.27±10.59	73.81±3.37	77.78±6.77	66.38±12.95	64.78±4.15	84.98±5.87	83.16±4.09
3	64.52± 8.95	59.30±6.13	76.38±7.73	76.14±8.63	76.40±9.62	75.90±0.56	84.45±1.20	85.98±1.00
4	13.98±13.35	25.12±11.58	70.58±10.46	30.85±20.12	43.01±18.54	57.33±12.97	53.07±31.33	57.31±19.60
5	56.39±8.02	74.56±6.44	82.27±2.60	84.52±3.04	87.25±2.90	83.82±2.52	90.37±3.83	88.78±2.88
6	1.25±2.50	22.98±22.98	58.40±27.49	5.71±8.06	11.30±14.63	55.91±0.26	3.51±5.62	2.56±3.14
7	37.55±20.08	52.74±7.77	43.84±15.93	63.08±11.24	61.10±10.54	31.94±20.15	75.57±14.05	78.01±10.80
8	84.10± 7.99	89.97±4.56	86.58±5.27	85.42±5.12	91.47±3.93	86.88±0.40	83.74±5.58	86.08±4.04
9	2.49±2.43	12.18±5.96	25.98±7.37	15.59±6.60	30.68±19.07	7.72±0.84	35.44±9.61	33.76±7.23
10	2.04± 1.98	1.24±1.17	1.85±1.52	2.35±1.94	2.28±3.68	4.92± 4.31	10.01±8.69	7.87±7.50
11	2.46± 3.04	56.33±32.15	31.27±13.76	14.23±13.68	11.48±19.22	8.57±29.28	16.17 ±16.72	24.72±23.18

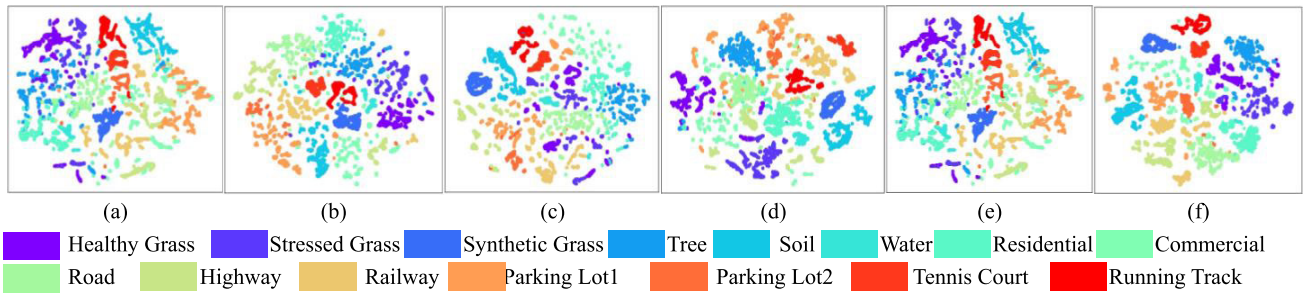


Fig. 15. t-SNE visualization on the Houston dataset. (a) EMP-SVM, (b) HT-CNN, (c) CNN, (d) MFT, (e) FMA, and (f) FMA^2 on the Houston dataset.

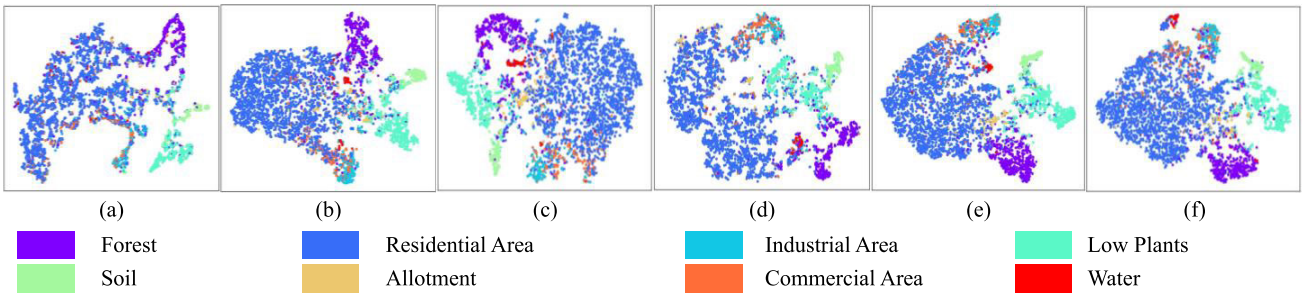


Fig. 16. t-SNE visualization on the Berlin dataset. (a) EMP-SVM, (b) HT-CNN, (c) CNN, (d) MFT, (e) FMA, and (f) FMA^2 on the Berlin dataset.

FMA selectively, and report the classification results including OA, AA, and K in Tables IV–VI.

It can be observed that using only $\boldsymbol{Z}_0$ in FMA, OA values are as low as 85.17%, 74.50%, and 76.25% on the Houston, Berlin, and MUUFL datasets, respectively, which demonstrates using the pretrained models for classification without fine-tuning of other information is ineffective. When both $\boldsymbol{S}_0$ and $\boldsymbol{D}_0$ are added to the single $\boldsymbol{Z}_0$, the classification accuracy is improved. With $\boldsymbol{S}_0$ or $\boldsymbol{D}_0$ only, OA decreased compared to the OA in the first row in Table IV. The results clearly reflect that the combination of $\boldsymbol{Z}_0$, $\boldsymbol{S}_0$, and $\boldsymbol{D}_0$ plays a crucial role in enhancing accuracy for multimodal RS image classification.

2) Each Modality of the Proposed FMA: There are two modalities in each dataset; the Houston and MUFFUL datasets both include the HSI and LiDAR, while the Berlin dataset consists of HSI and SAR. To validate the effectiveness of each modality of the proposed FMA, the F-CNN for multimodality classification is also displayed for comparison; the architecture of F-CNN is similar to the architecture in [36]. Figs. 12–14 show the classification results of different methods in terms of different multimodalities, where (a) and (b) represent the OA of each class of F-CNN or FMA with different modalities, respectively. It can be seen that the proposed FMA achieves the highest classification accuracy on a single modality, which demonstrates the effectiveness of the FMA.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

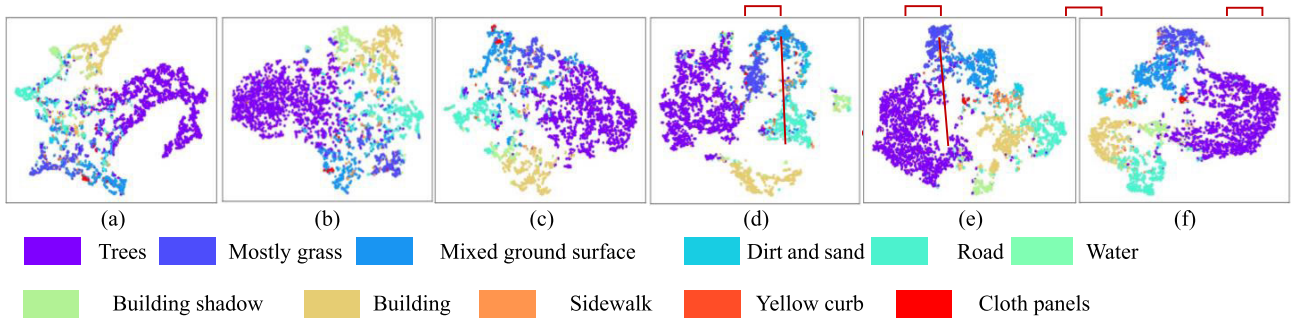


Fig. 17. t-SNE visualization on the MUUFL dataset. (a) EMP-SVM, (b) HT-CNN, (c) CNN, (d) MFT, (e) FMA, and (f) FMA^2 on the MUUFL dataset.

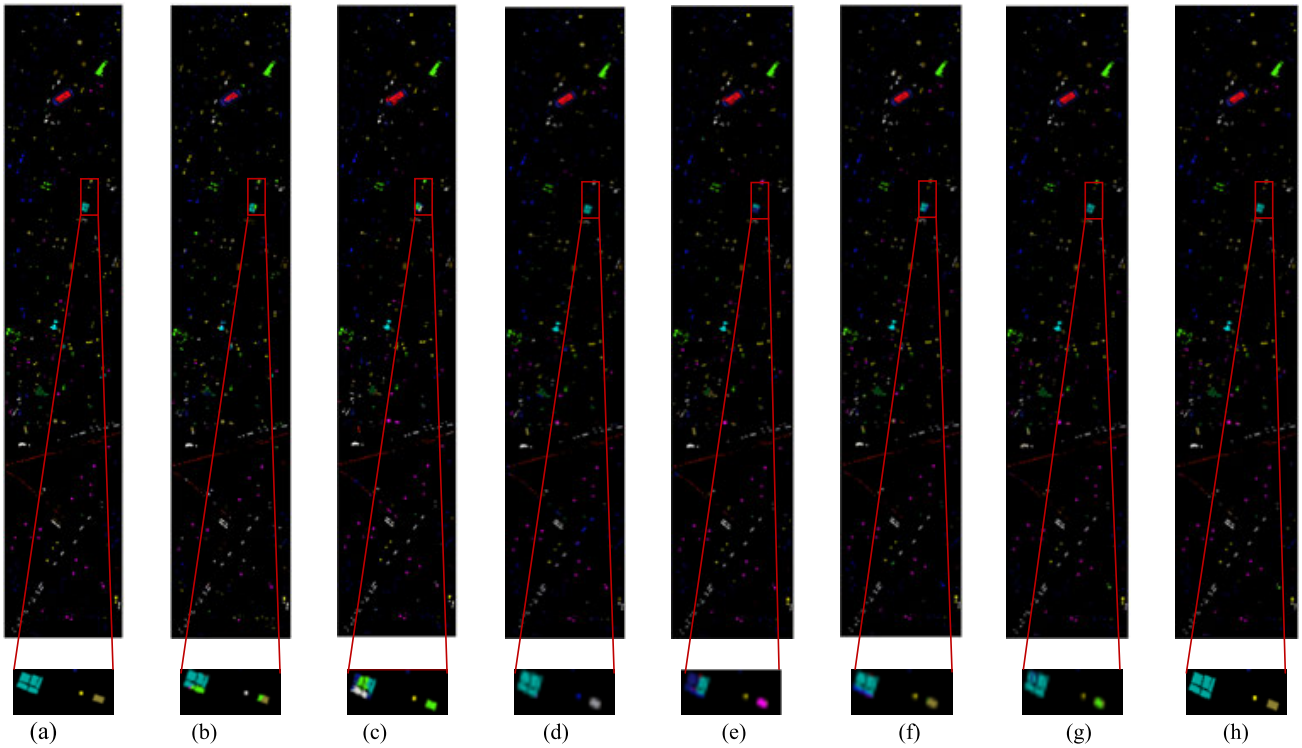


Fig. 18. Classification maps on the Houston dataset. (a) Ground-truth, (b) EMP-SVM, (c) HT-CNN, (d) SSFTT, (e) CNN, (f) MFT, (g) FMA, and (h) FMA^2.

As can be seen, compared to the F-CNN, the proposed FMA with different modalities outperforms F-CNN in terms of OA of each class using all multimodal RS images, which scatters in the outermost circle, especially for the Houston and MUUFL datasets. The results indicate that the proposed FMA achieves better performance with different modalities.

E. Classification Performance Analysis

Tables VII–IX present the experimental performance of the proposed FMA and FMA^2, and the compared methods EMP-SVM, HT-CNN, SSFTT, CNN, MFT, and CCRNet on the three multimodal RS images with 200 training samples. From the results, it can be seen that the proposed FMA and FMA^2 achieve the highest accuracy compared to other methods. Notably, when compared to the CNN, the proposed FMA exhibits the best OA with improvements of 2.66%, 2.08%, and 3.76% on the Houston, Berlin, and MUUFL datasets, respectively. When compared to the modality method based on the transformer-MFT, the proposed FMA^2, moreover, improves the classification results of the three datasets by 2.25%, 1.26%, and 3.66%, respectively. The experimental results reveal that the proposed FMA and FMA^2 based on the foundation models outperform the CNN and transformer-based classification methods, which are suitable for multimodal RS image classification.

F. Feature Visualization

To visually demonstrate the effectiveness of the proposed approaches (i.e., FMA and FMA^2) in classifying multimodal RS images, Figs. 15–17 present the t-distributed stochastic neighbor embedding (t-SNE) visualizations of various methods. t-SNE is able to showcase the distributions of output features obtained through various methods effectively by estimating a probability distribution of similarities in a low-dimensional space. Here, (e) and (f) are the feature visualization of the proposed methods, (a)–(d) are the compared methods. From Fig. 16, the distributions of output features in (e) and (f) show good clustering, such as the forest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

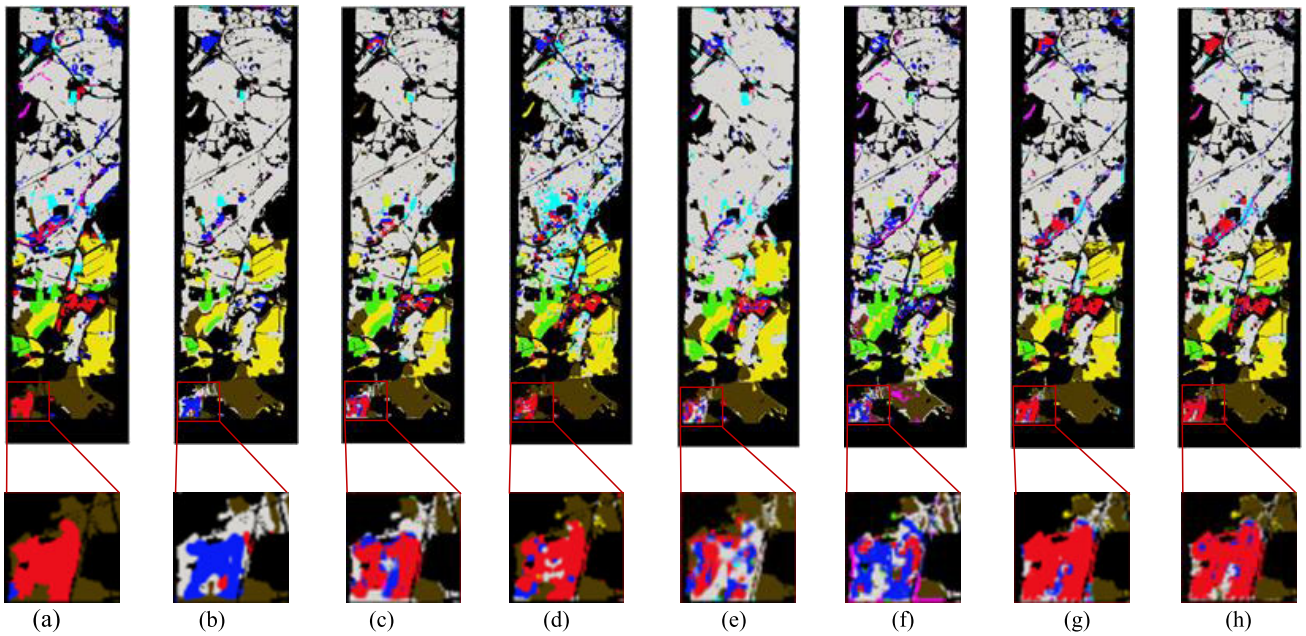


Fig. 19. Classification maps on the Berlin dataset. (a) Ground-truth, (b) EMP-SVM, (c) HT-CNN, (d) SSFTT, (e) CNN, (f) MFT, (g) FMA, and (h) FMA^2.

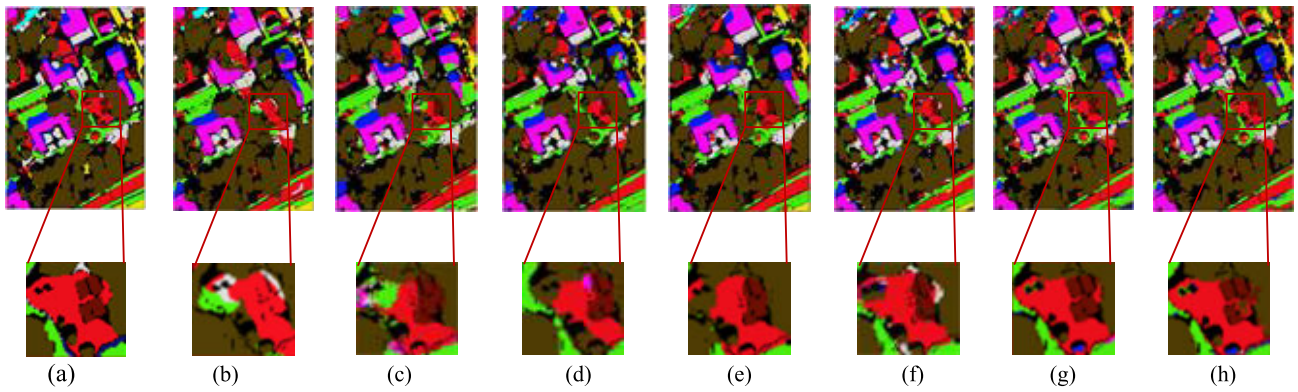


Fig. 20. Classification maps on the MUUFL dataset. (a) Ground-truth, (b) EMP-SVM, (c) HT-CNN, (d) SSFTT, (e) CNN, (f) MFT, (g) FMA, and (h) FMA^2.

TABLE X
TIME CONSUMPTION AND COMPUTATIONAL COMPLEXITY ON THE MULTIMODAL DATASETS

Datasets	Methods	HT-CNN	SSFTT	F-CNN	MFT	CCRNet	FMA	FMA^2
Houston	Computing Time (min.)	18.01	10.59	24.08	19.52	13.91	44.27	40.36
	Total parameters (M)	2.96	0.15	0.18	0.32	0.07	87.82	87.64
	Training parameters (M)	2.96	0.15	0.18	0.32	0.07	2.27	2.11
Berlin	Computing Time (min.)	41.13	15.39	34.67	32.83	93.86	107.89	101.41
	Total parameters (M)	2.95	0.15	0.18	0.40	0.09	87.84	87.67
	Training parameters (M)	2.95	0.15	0.18	0.40	0.09	2.34	2.15
MUUFL	Computing Time (min.)	13.11	13.34	16.30	14.11	23.41	36.64	32.27
	Total parameters (M)	2.95	0.15	0.18	0.26	0.06	87.80	87.61
	Training parameters (M)	2.95	0.15	0.18	0.26	0.06	2.27	2.08

class in purple color. Similarly, in Fig. 17, compared with the proposed methods, a within-class distance of the building class in yellow color in Fig. 17(a)–(d) is larger. The visual results indicate the efficacy of the proposed methods in increasing between-class distances and reducing within-class distances for improved class separability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

G. Classification Map

To facilitate visual comparison, Figs. 18–20 provide the obtained classification maps of various methods, including EMP-SVM, HT-CNN, SSFTT, CNN, MFT, FMA, and FMA^2. Upon observing these figures, the classification maps of the proposed FMA and FMA^2 exhibit fewer noisy scatter points on all the datasets. Taking the Berlin dataset as an example, in the bottom left corner of (g) in Fig. 19, the class industrial area with red color depicts fewer errors compared to other competing methods. The Houston and MUUFL datasets have similar visual results. All the figures demonstrate that the classification maps generated by the proposed FMA and FMA^2 appear to be smoother, which is consistent with the classification results.

H. Algorithm Complexity

Table X reports the computing time and total parameters of different deep learning-based methods (i.e., HT-CNN, SSFTT, F-CNN, MFT, CCRNet, FMA, and FMA^2) on the three datasets. The experiments are implemented on a computer with an Intel Core i7-11700 processor with 2.5 GHz and an NVIDIA GeForce RTX 3070 Ti graphical processing unit (GPU). It can be seen that, compared to other methods, the total numbers of parameters in FMA and FMA^2 are greater, and the reason is that the transformer used in the existing foundation model is usually larger than that in the HSI task, while the training parameters of the proposed FMA and FMA^2 are less than HT-CNN, which is a transfer strategy. Since the proposed methods are based on the pretrained foundation model, the computing time is longer than that of other compared methods. But the proposed methods achieve higher classification accuracy, and the increased computational cost is acceptable.

V. Conclusion

In this study, a foundation model-based method, i.e., FMA and FMA^2 are first proposed for multimodal RS data classification without training from scratch. Considering each RS modality has a spatial dimension and channel dimension, FMA explores two learnable modules, including a cross-spatial interaction module and a cross-channel interaction module, which extract multimodal-specific representations along different dimensions. Based on the FMA, FMA^2 proposes a coupling score function to solve the differences among multiple RS modalities effectively, leading to better classification performance. According to the experimental results, several key findings can be concluded as follows.

1) The transformer-based foundation model provides discriminative features for multimodal RS data classification. With the help of the foundation model, the intraclass distances are increased, and the interclass distances are reduced from the visualization results, which results in improved classification performance.

2) The proposed cross-spatial interaction module and cross-channel interaction module in FMA reserve the previous knowledge in foundation models, and better propagate the information from different RS multimodalities. Ablation studies show that both unimodality and multimodality achieve higher classification accuracy.

3) FMA^2 captures the relationships between different modalities (i.e., HSI, LiDAR, SAR) and increases the flexibility in aligning their representations in unimodality, leading to more effectiveness for multimodal RS data classification.

The proposed FMA and FMA^2 open up a new pathway in the classification of multimodal RS data. By integrating various RS modalities and leveraging the strengths of each modality, FMA and FMA^2 enable the exploration of more tasks (i.e., super resolution) by using multimodal RS data.

References

[1] C.-I. Chang, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification*. Berlin, Germany: Springer, 2003.

[2] T. Raj, F. H. Hashim, A. B. Huddin, M. F. Ibrahim, and A. Hussain, "A survey on LiDAR scanning mechanisms," *Electronics*, vol. 9, no. 5, p. 741, Apr. 2020.

[3] C.-A. Liu, Z.-X. Chen, Y. Shao, J.-S. Chen, T. Hasi, and H.-Z. Pan, "Research advances of SAR remote sensing for agriculture applications: A review," *J. Integrative Agricult.*, vol. 18, no. 3, pp. 506–525, Mar. 2019.

[4] J. Li, Y. Pei, S. Zhao, R. Xiao, X. Sang, and C. Zhang, "A review of remote sensing for environmental monitoring in China," *Remote Sens.*, vol. 12, no. 7, p. 1130, Apr. 2020.

[5] Y. Xiao and Q. Zhan, "A review of remote sensing applications in urban planning and management in China," in *Proc. Joint Urban Remote Sens. Event*, Shanghai, China, May 2009, pp. 1–5.

[6] J. Lee and L. Steele, "Military use of satellite communications, remote sensing, and global positioning systems in the war on terror," *J. Air L. Com.*, vol. 79, p. 69, Mar. 2014.

[7] A. E. Maxwell, T. A. Warner, and F. Fang, "Implementation of machine-learning classification in remote sensing: An applied review," *Int. J. Remote Sens.*, vol. 39, no. 9, pp. 2784–2817, May 2018.

[8] J. Chanussot, J. A. Benediktsson, and M. Fauvel, "Classification of remote sensing images from urban areas using a fuzzy possibilistic model," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 40–44, Jan. 2006.

[9] Y. Chen, K. Zhu, L. Zhu, X. He, P. Ghamisi, and J. A. Benediktsson, "Automatic design of convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7048–7066, Sep. 2019.

[10] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[11] A. Wehr and U. Lohr, "Airborne laser scanning—An introduction and overview," *ISPRS J. Photogramm. Remote Sens.*, vol. 54, nos. 2–3, pp. 68–82, Jul. 1999.

[12] B. Huang, Y. Li, X. Han, Y. Cui, W. Li, and R. Li, "Cloud removal from optical satellite imagery with SAR imagery using sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 5, pp. 1046–1050, May 2015.

[13] D. Stathakis and A. Vasilakos, "Comparison of computational intelligence based classification techniques for remotely sensed optical image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2305–2318, Aug. 2006.

[14] L. Huang and Y. Chen, "Dual-path Siamese CNN for hyperspectral image classification with limited training samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 518–522, Mar. 2021.

[15] P. Ghamisi, J. A. Benediktsson, and S. Phinn, "Land-cover classification using both hyperspectral and LiDAR data," *Int. J. Image Data Fusion*, vol. 6, no. 3, pp. 189–215, Jul. 2015.

[16] N. Yokoya et al., "Open data for global multimodal land use classification: Outcome of the 2017 IEEE GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 5, pp. 1363–1377, May 2018.

[17] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.

[18] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[19] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[20] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[21] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[22] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jul. 15, 2022, doi: 10.1109/TNNLS.2022.3189994.

[23] Y. Tay, M. Dehghani, D. Bahri, and D. Metzler, "Efficient transformers: A survey," 2020, *arXiv:2009.06732*.

[24] Y. Yuan, "On the power of foundation models," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 40519–40530.

[25] S. Bubeck et al., "Sparks of artificial general intelligence: Early experiments with GPT-4," 2023, *arXiv:2303.12712*.

[26] A. Kirillov et al., "Segment anything," 2023, *arXiv:2304.02643*.

[27] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[28] J. Wang, J. Li, Y. Shi, J. Lai, and X. Tan, "AM^{3}Net: Adaptive mutual-learning-based multimodal data fusion network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 8, pp. 5411–5426, Aug. 2022.

[29] M. Khodadadzadeh, J. Li, S. Prasad, and A. Plaza, "Fusion of hyperspectral and LiDAR remote sensing data using multiple feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2971–2983, Jun. 2015.

[30] B. Rasti, P. Ghamisi, and R. Gloaguen, "Hyperspectral and LiDAR fusion using extinction profiles and total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3997–4007, Jul. 2017.

[31] B. Rasti, P. Ghamisi, J. Plaza, and A. Plaza, "Fusion of hyperspectral and LiDAR data using sparse and low-rank component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6354–6365, Nov. 2017.

[32] D. Tuia, M. Volpi, M. Trolliet, and G. Camps-Valls, "Semisupervised manifold alignment of multimodal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 12, pp. 7708–7720, Dec. 2014.

[33] J. Peng, Y. Huang, W. Sun, N. Chen, Y. Ning, and Q. Du, "Domain adaptation in remote sensing image classification: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9842–9859, Nov. 2022.

[34] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3215677.

[35] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017.

[36] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.

[37] Y. Xu et al., "Advanced multi-sensor optical remote sensing for urban land use and land cover classification: Outcome of the 2018 IEEE GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1709–1724, Jun. 2019.

[38] N. Audebert, B. Le Saux, and S. Lefèvre, "Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 140, pp. 20–32, Jun. 2018.

[39] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[40] M. Zhang, F. Gao, T. Zhang, Y. Gan, J. Dong, and H. Yu, "Attention fusion of transformer-based and scale-based method for hyperspectral and LiDAR joint classification," *Remote Sens.*, vol. 15, no. 3, p. 650, Jan. 2023.

[41] M. Ma et al., "A multimodal hyper-fusion transformer for remote sensing image classification," *Inf. Fusion*, vol. 96, pp. 66–79, Aug. 2023.

[42] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[43] J. Zhou et al., "iBOT: Image BERT pre-training with online tokenizer," 2021, *arXiv:2111.07832*.

[44] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, no. 1, pp. 5485–5551, 2020.

[45] N. Dehouche, "Plagiarism in the age of massive generative pre-trained transformers (GPT-3)," *Ethics Sci. Environ. Politics*, vol. 21, pp. 17–23, Mar. 2021.

[46] Q. Zheng et al., "CodeGeeX: A pre-trained model for code generation with multilingual evaluations on HumanEval-X," 2023, *arXiv:2303.17568*.

[47] A. Ramesh et al., "Zero-shot text-to-image generation," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8821–8831.

[48] Y. Li et al., "Supervision exists everywhere: A data efficient contrastive language-image pre-training paradigm," 2021, *arXiv:2110.05208*.

[49] M. Caron et al., "Unsupervised learning of visual features by contrasting cluster assignments," in *Proc. Annu. Conf. Neural Inf. Process. Syst.*, 2020, pp. 1–13.

[50] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[51] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[52] X. He, Y. Chen, and P. Ghamisi, "Heterogeneous transfer learning for hyperspectral image classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3246–3263, May 2020.

[53] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3144158.

[54] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.



Xin He received the Ph.D. degree from the School of Electronics and Information Engineering, Harbin Institute of Technology (HIT), Harbin, China, in 2023.

She is currently an Assistant Professor with the School of Electronics and Information Engineering, HIT. Her research interests include remote sensing image processing based on deep learning methods.



Yushi Chen (Member, IEEE) received the Ph.D. degree from the Harbin Institute of Technology, Harbin, China, in 2008.

He is currently an Associate Professor with the School of Electronics and Information Engineering, Harbin Institute of Technology. His research interests include remote sensing data processing and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.



Lingbo Huang is currently pursuing the Ph.D. degree with the Department of Information Engineering, School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin, China.

His research interests include hyperspectral image classification, machine learning, and deep learning.



Danfeng Hong (Senior Member, IEEE) received the M.Sc. degree (summa cum laude) in computer vision from the College of Information Engineering, Qingdao University, Qingdao, China, in 2015, and the Dr.-Ing degree (summa cum laude) from the Signal Processing in Earth Observation (SiPEO), Technical University of Munich (TUM), Munich, Germany, in 2019.

Since 2022, he has been a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences (CAS), Beijing, China. Before joining CAS, he was a Research Scientist and led a Spectral Vision Working Group at the Remote Sensing Technology Institute (IMF), German Aerospace Center (DLR), Oberpfaffenhofen, Germany, from 2019 to 2021. He was also an Adjunct Scientist at GIPSA-lab, Grenoble INP, CNRS, University of Grenoble Alpes, Grenoble, France. His research interests include artificial intelligence, remote sensing big data analysis, multimodal interpretation, and their applications in Earth vision.

Dr. Hong was a recipient of the Best Reviewer Award of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2021 and 2022, and the Best Reviewer Award of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2022, the Jose Bioucas Dias Award for recognizing the outstanding paper at WHISPERS in 2021, the Remote Sensing Young Investigator Award in 2022, the IEEE GRSS Early Career Award in 2022, and a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) in 2022 and 2023. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), an Editorial Board Member of *Remote Sensing*, and an Editorial Advisory Board Member of the *ISPRS Journal of Photogrammetry and Remote Sensing*.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland at Baltimore, Baltimore, MD, USA, in 2000.

She is currently a Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral remote sensing image analysis and applications and machine learning.

Dr. Du was a member of the IEEE Periodicals Review and Advisory Committee and SPIE Publications Committee. She is a fellow of the SPIE-International Society for Optics and Photonics (SPIE). She received the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society (GRSS). She was the Co-Chair of the Data Fusion Technical Committee of the IEEE GRSS from 2009 to 2013; the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014; and the General Chair of the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing held at Shanghai, China, in 2012. She was an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, *Journal of Applied Remote Sensing*, and IEEE SIGNAL PROCESSING LETTERS. From 2016 to 2020, she was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:37:46 UTC from IEEE Xplore. Restrictions apply.